Viscoelastic Slender Jets

Parity argument (axis regularity + axisymmetry):

Velocity and pressure

Even powers for v_z, p and odd for v_r.

> **Compact notation:**
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> Primes are ∂_z:

Expanding v_z in powers of r:

$$v_z(r, z, t) = v_0(z, t) + \varepsilon^2 r^2 v_2(z, t) + \cdots,$$

Using continuity, v_r in the leading order is (see: Continuity equation in slender jet)

$$v_r(r, z, t) = -\frac{1}{2} v_0'(z, t)\, r - \frac{1}{4} \varepsilon^2 v_2'(z, t)\, r^3 + \cdots,$$

$$p(r, z, t) = p_0(z, t) + \varepsilon^2 r^2 p_2(z, t) + \cdots.$$

Stress

The Cauchy stress is

$$\boldsymbol{\sigma} = -p\boldsymbol{I} + 2\eta\boldsymbol{D} + \boldsymbol{\sigma}_p,$$

The momentum balance, free-surface traction balance, and shear-free condition use this $\boldsymbol{\sigma}$.

Expanding polymer stresses consistently with axisymmetry/regularity:

$$\sigma_{p,zz}(r, z, t) = \Sigma_{zz}(z, t) + O(\varepsilon^2 r^2), \qquad \sigma_{p,rr}(r, z, t) = \Sigma_{rr}(z, t) + O(\varepsilon^2 r^2),$$

$$\sigma_{p,rz}(r,z,t) = r\,S(z,t) + O(\varepsilon^2 r^3).$$

Hoop stress

Regularity also implies $\Sigma_{rr} = \Sigma_{\theta\theta}$ at leading order.

Governing equations

Axial momentum at $O(\varepsilon^0)$

Polymeric stress contribution:

$$(\nabla\cdot\boldsymbol{\sigma}_p)_z = \partial_z\sigma_{p,zz} + \frac{1}{r}\partial_r(r\sigma_{p,rz}).$$

Using $\sigma_{p,zz} \approx \Sigma_{zz}(z,t)$ and $\sigma_{p,rz} \approx rS(z,t)$,

$$\frac{1}{r}\partial_r(r\sigma_{p,rz}) = \frac{1}{r}\partial_r(r^2 S) = 2S.$$

Thus the r^0 coefficient of the axial equation becomes (why factor 4? See: Laplacian in axisymmetric slender jet. For full derivation of LHS, see LHS of slender jet momentum)

$$\rho\left(\partial_t v_0 + v_0 v_0'\right) = -p_0' + \eta(4v_2 + v_0'') + \Sigma_{zz}' + 2S + O(\varepsilon^2).$$

Kinematic BC:

See Continuity equation in slender jet for details

$$\partial_t(h^2) + \partial_z(h^2 v_0) = 0,$$

Leading-order stress BCs with polymer normal + shear stresses

See: Full free surface stress balance **for details.**

Normal traction (leading order, $n \approx e_r$):

$$p_0 = \frac{\gamma}{h} - \eta v_0' + \Sigma_{rr} + O(\varepsilon^2).$$

Tangential traction (leading order, $t \approx e_z$, $n \approx e_r$):

$$\underbrace{\eta\Big(-3v_0'h' - \frac{1}{2}v_0''h + 2v_2h\Big)}_{\text{Newtonian part}} + \underbrace{\sigma_{p,rz}(h)}_{\approx hS} + \underbrace{(\Sigma_{rr} - \Sigma_{zz})h'}_{=-\Delta\Sigma\, h'} = O(\varepsilon^2),$$

where polymeric "tensile" normal-stress difference is

$$\Delta\Sigma \equiv \Sigma_{zz} - \Sigma_{rr}.$$

Combining

Using the tangential stress balance to solve for v_2 (substituting $O(\varepsilon^2) \to 0$):

$$2\eta v_2 h = 3\eta v_0'h' + \frac{\eta}{2}v_0''h - \sigma_{p,rz}(h) + \Delta\Sigma\, h'.$$

Next use the normal stress BC for p_0:

$$p_0' = (\gamma/h)' - \eta v_0'' + \Sigma_{rr}'.$$

Now substitute into the axial momentum equation and using $\sigma_{p,rz} \approx rS(z,t)$:

$$\rho\left(\partial_t v_0 + v_0 v_0'\right) = -\left(\frac{\gamma}{h}\right)' + 3\eta\frac{(h^2 v_0')'}{h^2} + \frac{(h^2\Delta\Sigma)'}{h^2} + O(\varepsilon^2),$$

with $\Delta\Sigma = \Sigma_{zz} - \Sigma_{rr}$. Here, 3 is the Trouton ratio.

$\Sigma_{rr} = \Sigma_{\theta\theta}$ at leading order (so "hoop" is already accounted for).

> **Important**
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> What happened to the shear stress $\sigma_{p,rz}$? It enters both (i) the axial momentum equation through $(\nabla \cdot \sigma_p)_z$, and (ii) the tangential traction condition $t \cdot \sigma \cdot n = 0$. Those two appearances will cancel.

Conservative form

Continuity (see: Continuity equation in slender jet for details):

$$\boxed{\partial_t h^2 + \partial_z (h^2 v_0) = 0}$$

Momentum equation:

$$\rho(\partial_t v_0 + v_0 v_0') = -\left(\frac{\gamma}{h}\right)' + 3\eta \frac{(h^2 v_0')'}{h^2} + \frac{(h^2 \Delta\Sigma)'}{h^2}.$$

Multiply by h^2:

$$\rho h^2(\partial_t v_0 + v_0 v_0') = -h^2 \left(\frac{\gamma}{h}\right)' + 3\eta \big(h^2 v_0'\big)' + \big(h^2 \Delta\Sigma\big)'.$$

Using $h^2(1/h)' = -h'$:

$$\rho h^2(\partial_t v_0 + v_0 v_0') = \gamma \Big(h\Big)' + 3\eta \big(h^2 v_0'\big)' + \big(h^2 \Delta\Sigma\big)'.$$

On the left hand side:

$$\partial_t(h^2 v_0) + (h^2 v_0^2)' = h^2(v_t + v v') \quad \text{if (by continuity equation)} \quad \partial_t(h^2) + (v_0 h$$

The conservative momentum equation is

$$\rho \left(\partial_t(h^2 v_0) + (h^2 v_0^2)'\right) = \left(\gamma h + 3\eta h^2 v_0' + h^2 \Delta\Sigma\right)'.$$

$$\boxed{\rho \left(\partial_t(h^2 v_0) + \partial_z(h^2 v_0^2)\right) = \partial_z \left[\gamma h + 3\eta h^2 \partial_z v_0 + h^2 \left(\sigma_{p,zz} - \sigma_{p,rr}\right)\right].}$$